Crossdem: Comparing Politicians' Rhetoric With Democracy Indicators

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Comparison between politicians' speeches and V-Dem democracy indices. After building a corpus of public speeches from Italian Prime Minister, U.S. Presidents and French Prime Ministers and Presidents, NLP analyses will be performed, from text complexity to hate speech recognition. The ensuing results will be compared with V-Dem democracy indices to shed light on possible correlations. Lastly, a transformer model will be fine-tuned to interpret political language and classify hate speech.

CCS Concepts: • **Computing methodologies → Natural language processing**; • **Social and professional topics → Hate speech**; **Political speech**.

Additional Key Words and Phrases: AI, NLP, Politics, Speeches

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1 INTRODUCTION

Politicians are some of the most influential people in human society: from the Greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to steer society. Shapiro and Page (1984) [19] showed how nearly all 20^{th} century U.S. Presidents have tried to lead public opinion, with varying degrees of success. As societies became more democratic, so grew the leaders' responsibility to convince potential followers that their policies are beneficial. Thus, speeches play a vital role in the functioning of society [3]: leaders depend on the verbal power to persuade people [7].

Moreover, democratic stability depends on citizens on the losing side accepting election outcomes. Clayton et al. [9] evaluated the effects of exposure to multiple statements from (at the time) former president Donald Trump attacking the legitimacy of the 2020 US presidential election. They found no evidence indicating that elites can erode democratic norms easily or that the effects of norm violations are uniform across the entire population. However, elite rhetoric can shape normative beliefs in core democratic values such as confidence in elections and support for peaceful transfers of power. This was especially evident amongst people who approved of Trump's performance in office.

Having established the importance of speeches, we can differentiate between two types: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social network posts. This last category is especially interesting since it affords politicians a direct and high-volume communication channel to their potential followers: U.S. President Donald Trump tweeted 57,000 times between May 2009 and January 2021 alone [16], while Italy's Prime Minister Giorgia Meloni totalled more than eighty million social network interactions in 2025 [25].

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On the other hand, internal speeches are especially easy to access, being already aggregated in public databases, yet their technical and codified nature makes their interpretation harder for both human and automated agents. Moreover, until 1985 Italian stenographic documentation did not report speeches word for word, but rather translated them into a "language of the parliament" [13, 18].

2 SOCIAL NETWORKS

Broadly speaking, the goal of this mini-project is to gather a corpus of politicians' speeches and perform linguistic analyses on it. The idea is to start from standard NLP tasks such as word frequency and text complexity, to then move to classification tasks such as sentiment analysis and hate speech detection.

This objective is why I ultimately decided to gather a corpus of *external speeches* rather than internal ones. Large language models are trained with tokenizers, and the resulting token distribution is highly imbalanced: Chung et al. [8] did a controlled study that scaled the vocabulary of the language model from 24K to 196K while holding data, computation, and optimization unchanged. They discovered that models are disproportionately optimized on the part of language that appears most often, meaning they are way better at understanding common words and patterns. This translates to the fact that internal speeches are less understandable for an AI agent.

As previously stated, external speeches include social network communications. While television is still the first source of information for the majority of the population, online platforms are steadily growing and are already the preferred media for young people [10, 17]. There is evidence of an increase in political participation due to social media usage, as well as risks for the functioning of democracy [2, 15], and there have been attempts to predict election results with social media data analyses. Rita et al. measure sentiment polarity on Twitter, and conclude that tweets' sentiment is not a reliable election results predictor. Additionally, results also show that it is impossible to state that social media impacts voting decisions [21].

Opposite results can be seen in Belcastro et al.'s [5] 2022 study: a real-time analysis was carried out during the 2020 US presidential election campaign, correctly identifying the leading candidate before Election Day in 10 out of 11 swing states.

Silva et al. [23] managed to match tweets against parliamentary speeches to measure politicians' sentiment on a specific topic. They find that parliament members who participate less in parliamentary debate tend to have larger differences with their party on Twitter, suggesting that a certain level of self-censoring is taking place in the parliamentary arena.

That being said, X is just one channel among many used to connect with the voters: official messages to the nation, interviews, and public speeches are still powerful tools, and are often made publicly available in video format by the politicians themselves on their Facebook page and Youtube channel.

3 PROJECT GOAL

Let's now define the goal of the mini-project.

I want to compare the rhetoric between the first and the current[1] Italian Prime Minister, namely Alcide De Gasperi and Giorgia Meloni. The rhetoric will be compared using natural language processing techniques, such as measuring textual complexity and hate speech classification. Only public speeches will be taken, thus leaving out all parliamentary speeches.

[1] At the moment of writing: May 2026.

I also want to produce a proof-of-concept for a speech-to-text pipeline that, given a set of URLs, each of which a Youtube's permalink, transcribe the video into text.

The first step will be gathering two corpora of public speeches, one for each PM. In the case of Giorgia Meloni, I want to compile the corpus by transcribing video interviews and public talks available on her official Youtube channel.

Then, after having gathered the corpora into suitable machine-readable data (such as csv), I will pre-process them with procedures such as tokenization, stemming and lemmatization. Lastly, I will compute word frequency and text complexity measures, as well as hate speech classification and sentiment analysis.

4 BUILDING THE CORPUS

4.1 Alcide De Gasperi's Corpus

I would like to thank Sara Tonelli for providing me with the De Gasperi's Corpus [24], a collection of Alcide De Gasperi's public documents with gold and silver annotation.

The corpus is a collection of 2,762 documents issued between 1901 and 1954, formatted into XML files which include metadata that covers not only the title, the date and the place of publication, but also key-concepts automatically extracted from each text (with the corresponding relevance score) and genre labels manually assigned by domain experts. Furthermore, the release includes silver annotation for lemma, part of speech, person names and place names with associated coordinates in a CoNLL-like format.

An example of the XML formatting can be seen at listing 1. Thanks to the *<genres>* tag, I was able to extract 474 public speeches, each of which with a location, a precise date and a list of keywords.

Listing 1. An example of how a single speech in De Gasperi's Corpus is formatted in XML

```xml
<?xml version="1.0" encoding="UTF-8" standalone="no"?>
<document id="I.doc_7">
   <publication_date>1901-11-26</publication_date>
   <publication_place>
      <place_name>Trento</place_name>
      <place_latitude>46.0664228</place_latitude>
      <place_longitude>11.1257601</place_longitude>
   </publication_place>
   <keywords>
      <keyword score="39.16">first keyword</keyword>
      ...
      <keyword score="5.59">last keyword</keyword>
   </keywords>
   <genres>
      <genre>Speech Type</genre>
   </genres>
   <text>The speech itself if stored between these brackets</text>
</document>
```

4.2 Giorgia Meloni's Corpus

My objective was to build Meloni's corpus by transcribing videos of her public speeches, such as talks and interviews. Fortunately, there is an unofficial Youtube channel (that I reached from the Prime Minister's official website) which aggregates more than four thousand videos of her public appearances. The channel is called "Giorgia Meloni News"[2].

[2]Giorgia Meloni News: https://www.youtube.com/@GiorgiaMeloniTv

Given a Youtube URL, I can use Python's library *yt-dlp* to retrieve the video's metadata and download its audio content in mp3 format, as shown in listing 2. I had to manually pass the cookies taken from my web-browser, and I used some extra commands to prevent Youtube to block the requests due to suspicious activity.

Then, the mp3 file just retrieved gets injected into OpenAI's Whisper library, which uses an encoder-decoder Transformer to transcribe it into text, as shown in listing 3. The model I used is the *"medium"*[3], which at 5 GB fits within the total 6 GB of VRAM available to my RTX 4050 GPU. The model was instructed to predict Italian.

The difference in precision between the models "tiny" (which requires less than 1 GB of VRAM) and "medium" is quite high, as can be seen in the following pieces of text that have been generated from the same audio file:

- **Model tiny:** "iamo con se è beruto fuori da questa rio neone sotto il profiro tecnico per quanto riguarda la franha e come il governo un tino era ad essere vicino alla popolazione di Nishenia."
- **Model medium:** "Diciamo cosa è venuto fuori da questa riunione sotto il profilo tecnico per quanto riguarda la frana e come il governo continuerà ad essere vicino alla popolazione diniscemeca."

The last step of the pipeline saves the transcription into a csv file, along with the extracted metadata. Each file has the following fields: *politician* ("meloni" in this case), *historical_date* (the upload date of the video), *location* and *tags* (extracted from the metadata if available, empty strings otherwise), *description* and *title* of the video, *url* which stores the permalink of the video itself, and lastly *text* which holds the whole transcription.

Each video is processed immediately after being retrieved, and until it has been saved in a csv file the program does not try to fetch the next one. This is done for two reasons: first, to prevent *yt-dlp* from sending requests too close to each other, which could result in Youtube denying them due to suspicious activity. Second, even if the pipeline halts for any reason, the videos processed up to that point will not be lost.

Listing 2. Downloading audio and metadata from a Youtube video

```
result = subprocess.run([
    "yt-dlp",
    "--print-json",
    "-x",                               # download audio only
    "--audio-format", "mp3",
    "--cookies", "yt_cookies.txt",      # manual cookies, extract with browser extension
    "--sleep-requests", "2",            # Sleep 2s between requests
    "--sleep-interval", "5",            # Sleep 5s between downloads
    "--max-sleep-interval", "15",       # Randomize up to 15s
    "--limit-rate", "5M",               # Throttle to 5MB/s (mimics streaming)
    "-o", audio_file.replace('.mp3', ''),
    video_url
], capture_output=True, text=True, check=True)
```

Listing 3. Transcribe the audio file into text using Whisper

```
subprocess.run([
    "whisper",
    "--language",         lang_code,     # "Italian" chosen
    "--word_timestamps", "True",
    "--model",            model_name,    # model "medium" chosen
    "--output_dir",       output_dir,
    "--device",           "cuda",        # ensures the model gets loaded in GPU
    audio_file
```

[3]Whisper model sizes available are tiny, base, medium and large.

```
], check=True)
```

5 LINGUISTIC ANALYSIS

Sawicki et al. 2023's survey [22] tells us that English is by far the most represented language in the field of natural language processing. Fortunately, French and Italian are still in the top eight (figure 1), so finding libraries already capable of handling them should be easy enough. It is worth noting that results could be biased since the authors looked for papers written in English.

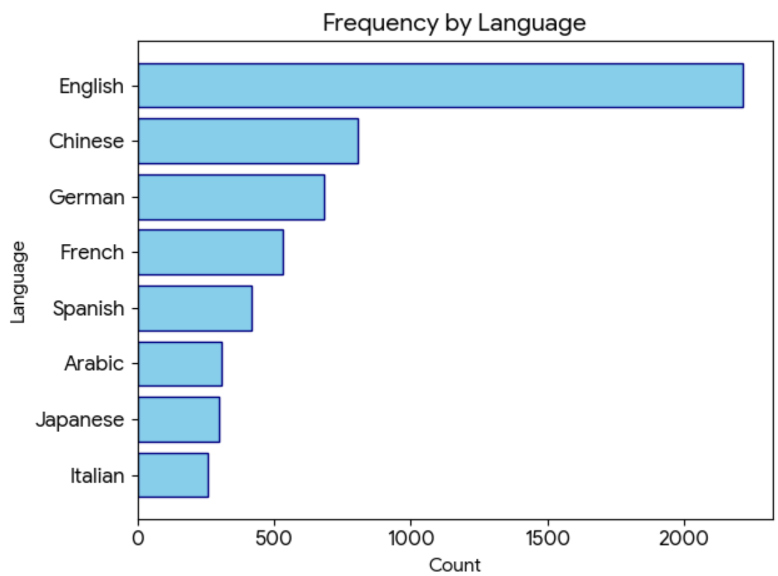

Fig. 1. Most popular languages in the field of Natural Language Processing [22]

The pipeline for pre-processing is quite standard and involves, among other steps, removing stop words, part-of-speech tagging, tokenization and lemmatization, as well as building text embeddings.

FastText[4] is an open-source, free, lightweight library that allows users to learn text representations and text classifiers; it gained significant popularity due to its performance in computing text embeddings.

LLM2Vec [4] is a recipe to convert decoder-only LLMs into text encoders so that they can be used to generate state-of-the-art text embeddings.

Stanza[5] is a collection of accurate and efficient tools for the linguistic analysis of many human languages. Starting from raw text, Stanza divides it into sentences and words, and can then recognize parts of speech and entities, perform syntactic analysis, and more. Stanza brings state-of-the-art NLP models to the languages of your choosing.

spaCy[6] is another well-known and powerful NLP library for Python that supports multiple languages. It also supports tokenization, part-of-speech tagging, dependency parsing, sentence segmentation, text classification, and

[4]FastText is a lightweight library for efficient text classification and representation learning: https://fasttext.cc/.
[5]Stanza – A Python NLP Package for Many Human Languages: https://stanfordnlp.github.io/stanza/index.html.
[6]spaCy, Industrial-Strength Natural Language Processing: https://spacy.io/.

lemmatization. It supports multi-task learning with pretrained transformers, includes pretrained word vectors, supports custom models in PyTorch and TensorFlow, and has built-in syntax visualizers.

simplemma[7] is also worth mentioning as a lightweight Python lemmatizer.

textacy[8] is a Python library for performing a variety of natural language processing (NLP) tasks. It is built on spaCy and intended to be used both before (pre-processing step) and after (NLP step) it. It assesses text complexity with the Flesch-Kincaid readability test and text diversity with the Type-Token Ratio.

Text complexity can also be measured with **TRUNAJOD**[9], another spaCy-based Python library that implements synonym overlap, coherence between sentences, and other metrics.

Regarding populism and hate speech classification, there are many transformer-based models available from **Hugging Face**[10]. Some examples follow:

- English: https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-english;
- French: https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-french;
- Italian: https://huggingface.co/Hate-speech-CNERG/dehatebert-mono-italian.

There are transformer models based on RoBERTa for both hate speech and populism classification. For example, the following is a 0.4 billion parameter model based on Trump speeches: https://huggingface.co/coastalcph/roberta-large-ft-trump-populism.

Hate speech can also be detected with the Python library **HateSonar**[11], which uses a BERT-based model.

5.1 Speech To Text

Since there are plenty of video and audio speeches publicly available, they could be integrated into the corpus after transcribing them into text. This can be achieved with libraries such as **SpeechRecognition**[12], which supports several different engines and APIs both online and offline, or OpenAI's **Whisper**[13] library. Whisper is a Transformer sequence-to-sequence model trained on various speech processing tasks, including multilingual speech recognition, speech translation, spoken language identification, and voice activity detection. It comes in six different model sizes; the "small" model requires approximately 2GB of VRAM, making it easy to run locally.

Whisper's performance varies depending on the language. Italian is the fourth best-performing language with a WER[14] of 5.5%. English follows at 9.3% and French at 10.8% [20]. The worst-performing language is Albanian with a WER of 55.7%.

Whisper can be used both in Python scripts and via a command-line interface.

[7] simplemma is a lightweight toolkit for multilingual lemmatization and language detection: https://pypi.org/project/simplemma/.
[8] textacy: NLP, before and after spaCy. https://pypi.org/project/textacy/.
[9] TRUNAJOD: A text complexity library for text analysis built on spaCy. https://trunajod20.readthedocs.io/en/latest/.
[10] Hugging Face, Inc., is an American company based in New York City that develops computational tools for building applications using machine learning. Its transformers library, built for natural language processing applications, and its platform allow users to share machine learning models and datasets and showcase their work. https://huggingface.co/
[11] HateSonar: Hate Speech Detection. https://pypi.org/project/hatesonar/.
[12] SpeechRecognition, a Python library for performing speech recognition with support for several engines and APIs, online and offline. https://pypi.org/project/SpeechRecognition/.
[13] Whisper is a general-purpose speech recognition model. https://github.com/openai/whisper.
[14] Word Error Rate (WER) is a common metric for the performance of speech recognition or machine translation systems. It is measured as an error rate percentage for compared sentences.

6 DEMOCRACY INDEX

There are plenty of sources for measuring democracy levels. Our World in Data mainly uses six: Varieties of Democracy (V-Dem), the Lexical Index of Electoral Democracy (LIED) by Skaaning et al. (2015), Freedom House's (FH) Freedom in the World index, the Bertelsmann Transformation Index (BTI) by the Bertelsmann Foundation, the Economist Intelligence Unit's (EIU) Democracy Index, and Polity by the Center for Systemic Peace [11].

Out of the six mentioned datasets, only V-Dem and LIED have data spanning between 1946 and 2025. However, V-Dem offers more granularity and is generally considered a more complete and robust dataset. It is the best choice if we are interested in both large and small differences in varieties of democracy far into the past, or if we want to use country experts to measure characteristics of political systems that are difficult to observe. These experts are anonymous and are primarily academics or members of the media and civil society. They are also often nationals or residents of the country they assess; therefore, they know its political system well and can evaluate aspects that are difficult to observe. V-Dem's own team of researchers supplements these expert evaluations [12].

6.1 Varieties of Democracy Dataset

All data is available for download at https://v-dem.net/data/the-v-dem-dataset/ or can be accessed directly as an R package.

V-Dem distinguishes between *indicators*—lower-level data that is often not normalized—and *indices*, which aggregate indicators and are normalized on a scale between zero and one.

There are four datasets: *Coder-Level*, with raw data and 273 indicators intended for the experts; *CD Country-Date*, with day-to-day granularity, 531 indicators, and 95 aggregated indices; *CY Country-Year*, with year-level granularity, 5 indices, 93 sub-indices, and 179 indicators; and lastly, *CY Full + Others*, with year-level granularity, 531 indicators, 251 indices, and 62 external indicators.

I must use the *CD Country-Date* dataset (version 16, released in March 2026) since, to keep track of Italian governments, year-level granularity is not enough. This is evident from Figure 2: half of Italy's governments lasted less than a year.

The dataset covers Italy (ID 82) between 1861 and 2025, France (ID 76) between 1789 and 2025, and the U.S. (ID 20) between 1789 and 2025.

Useful identifiers include: `country_id`, unique for each country; `country_text_id`, ISO 3166-1 alpha-2 country codes (e.g., *IT*, *FR*); `historical_date`, in *YYYY-MM-DD* format, used for election-date-specific variables (all other variables use January 1st); and `gapstart` and `gapend`, representing time periods where no data is available.

Democracy Indices are the highest-level indices, each normalized to a scale between zero and one unless otherwise specified: `v2x_polyarchy`, the *electoral* democracy index (aggregates `v2x_elecoff`, `v2xel_frefair`, `v2x_frassoc_thick`, `v2x_suffr`, `v2x_freexp_altinf`); `v2x_libdem`, the *liberal* democracy index (aggregates `v2x_polyarchy`, `v2x_liberal`); `v2x_partipdem`, the *participatory* democracy index (aggregates `v2x_polyarchy`, `v2x_partip`); `v2x_delibdem`, the *deliberative* democracy index focusing on process, respectful dialogue, and the common good (aggregates `v2x_polyarchy`, `v2xdl_delib`); and `v2x_egaldem`, the *egalitarian* democracy index considering social status, inequalities, and resource distribution (aggregates `v2x_polyarchy`, `v2x_egal`).

Components of Democracy Indices consist of second-level indices and subcomponents: `v2x_freexp_altinf`, freedom of expression and alternative information (aggregates *v2mecenefm*, *v2meharjrn*, *v2meslfcen*, *v2xcl_disc*, *v2clacfree*, *v2mebias*, *v2mecrit*, *v2merange*); `v2x_frassoc_thick`, freedom of association for parties and civil society organizations

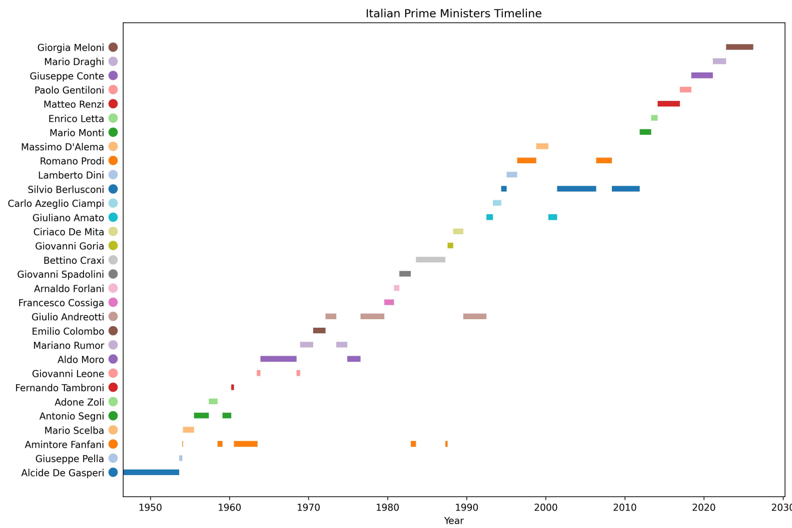

Fig. 2. Italian Prime Ministers for each Italian Republic, from the first (1946) to the latest (2025).

(aggregates *v2psparban*, *v2psbars*, *v2psoppaut*, *v2elmulpar*, *v2cseeorgs*, *v2csreprss*, `v2x_elecreg`); `v2x_suffr`, the percentage of adult suffrage (aggregates *v2elsuffrage*); `v2xel_frefair`, the quality of free and fair elections (aggregates *v2elembaut*, *v2elembcap*, *v2elrgstry*, *v2elvotbuy*, *v2elirreg*, *v2elintim*, *v2elpeace*, *v2elfrfair*, `v2x_elecreg`); `v2xcl_rol`, rule of law, transparency, and fair enforcement (aggregates *v2clrspct*, *v2cltrnslw*, *v2cltort*, *v2clkill*, *v2clrelig*, *v2clfmove*, `v2xcl_dmove`, `v2xcl_slave`, `v2xcl_acjst`, `v2xcl_prpty`); `v2x_jucon`, executive law compliance and judiciary independence (aggregates *v2exrescon*, *v2jucomp*, *v2juhccomp*, *v2juhcind*, *v2juncind*); `v2xlg_legcon`, executive external scrutiny (aggregates *v2lgqstexp*, *v2lgotovst*, *v2lginvstp*, *v2lgoppart*); `v2x_partip`, individual participation (aggregates `v2x_cspart`, `v2xdd_dd`, `v2xel_locelec`, `v2xel_regelec`); `v2x_cspart`, participation in associations like labor unions or NGOs (aggregates *v2pscnslnl*, *v2cscnsult*, *v2csprtcpt*, *v2csgender*); `v2xdd_dd`, the direct vote index for referendums and ballots (aggregates *v2ddlexci* through *v2ddthreci*); `v2xel_regelec` and `v2xel_locelec`, regional and local government elections; `v2xeg_eqprotec`, equal rights across social groups; `v2xeg_eqaccess`, equality of accessing power; and `v2xeg_eqdr`, equality of resource distribution (aggregates *v2dlencmps*, *v2dlunivl*, *v2peedueq*, *v2pehealth*).

7 MODEL FINE-TUNING

Classification tasks can be performed, broadly speaking, in two ways: the first leverages common machine learning and statistical tools, such as logistic regression; the second approach uses Transformer models (commonly referred to today as "AI"). A traditional approach might involve using TF-IDF and logistic regression with the Scikit-Learn library (example code at 4).

Listing 4. Scikit-Learn Logistic Regression for classification task

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline

X_train, X_test, y_train, y_test = train_test_split(df['text'], df['label'], test_size=0.2)
```

```python

text_clf = Pipeline([
    ('tfidf', TfidfVectorizer(stop_words='english', ngram_range=(1, 2))),
    ('clf', LogisticRegression())
])
```

On the other hand, Transformers are far superior in terms of accuracy for textual analysis, although they are much more resource-demanding. An example would be using Hugging Face's `transformers` library to fine-tune a pre-trained model (example code at 5).

Listing 5. Hugging Face Trainer for fine-tuning

```python
from transformers import AutoModelForSequenceClassification, AutoTokenizer, Trainer, TrainingArguments

trainer = Trainer(
    model=AutoModelForSequenceClassification.from_pretrained(model_name, num_labels=2),
    args=training_args,
    train_dataset=tokenized_datasets["train"],
    eval_dataset=tokenized_datasets["test"],
)

trainer.train()
```

Regarding the literature on the subject, Card et al. [6] fine-tuned a pre-trained RoBERTa model on congressional speeches in a self-supervised fashion and then further fine-tuned it as a classifier using annotated examples. They achieved 65% accuracy in classifying Congressional Record entries as being pro-immigration, anti-immigration, or neutral.

Albladi et al. [1] surveyed the field of NLP sentiment analysis and found that RoBERTa consistently outperforms other methods in multi-classification tasks. Their results revealed a clear correlation between dataset size and improved model performance, with larger datasets significantly increasing both accuracy and F1 scores across all models. For example, GPT-3.5 reached accuracies of 0.9694, 0.9784, 0.9797, and 0.9875 for various dataset sizes (e.g., 500, 1000, and 2500). However, LLaMA-2 13B emerged as the most balanced option for NLP tasks, striking an optimal balance between computational efficiency, performance, and adaptability. This model can be fine-tuned on freely available GPUs, such as the T4 on Google Colab, using techniques like QLoRA, which makes it an attractive choice for researchers and developers.

Li et al. [14] found that LS-LLaMA substantially outperforms LLMs ten times its size in text classification tasks and demonstrates consistent improvements compared to robust baselines like BERT-Large and RoBERTa-Large.

Considering all of this, it would be interesting to fine-tune a Mistral 7B model, which is claimed to perform better than LLaMA 7B and LLaMA 13B across all benchmarks (Figure 3).

8 FEASIBILITY

The project can be completed within the three-year duration of the PhD program. One year is expected to be dedicated to completing the dataset, leaving sufficient time to conduct the ensuing textual analyses.

Furthermore, the majority of the work required for the Italian corpora will be completed in the coming months as part of my Master's thesis project.

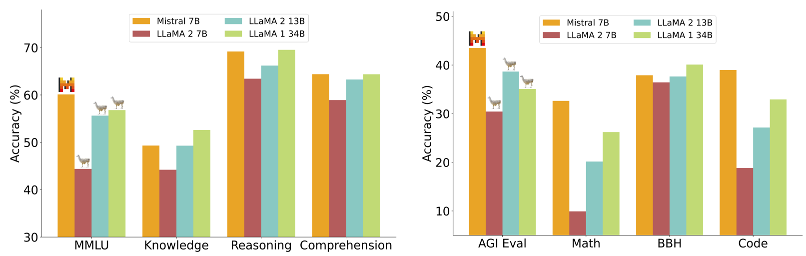

Fig. 3. Mistral 7B consistently outperforms LLaMA on all tasks. Source: Mistral.

This work will tie neatly to my career progression since I changed my specialization from cybersecurity to a more data analysis-oriented path. I also discovered an interest in AI, and specifically natural language processing through transformer models, which this PhD will allow me to explore.

REFERENCES

[1] Aish Albladi, Minarul Islam, and Cheryl Seals. 2025. Sentiment Analysis of Twitter Data Using NLP Models: A Comprehensive Review. *IEEE Access* 13 (2025), 30444–30468. https://doi.org/10.1109/ACCESS.2025.3541494

[2] Eran Amsalem and Alon Zoizner. 2023. Do people learn about politics on social media? A meta-analysis of 76 studies. *Journal of Communication* 73, 1 (Feb. 2023), 3–13. https://doi.org/10.1093/joc/jqac034

[3] Adrian Beard. 2000. *The Language of Politics*. Routledge. https://newuniversityinexileconsortium.org/wp-content/uploads/2022/02/the-language-of-politics.pdf Google-Books-ID: XpqSmh1pjsEC.

[4] Parishad BehnamGhader, Vaibhav Adlakha, Marius Mosbach, Dzmitry Bahdanau, Nicolas Chapados, and Siva Reddy. 2024. LLM2Vec: Large Language Models Are Secretly Powerful Text Encoders. https://doi.org/10.48550/arXiv.2404.05961 arXiv:2404.05961 [cs].

[5] Loris Belcastro, Francesco Branda, Riccardo Cantini, Fabrizio Marozzo, Domenico Talia, and Paolo Trunfio. 2022. Analyzing voter behavior on social media during the 2020 US presidential election campaign. *Social Network Analysis and Mining* 12, 1 (July 2022), 83. https://doi.org/10.1007/s13278-022-00913-9

[6] Dallas Card, Serina Chang, Chris Becker, Julia Mendelsohn, Rob Voigt, Leah Boustan, Ran Abramitzky, and Dan Jurafsky. 2022. Computational analysis of 140 years of US political speeches reveals more positive but increasingly polarized framing of immigration. *Proceedings of the National Academy of Sciences* 119, 31 (Aug. 2022), e2120510119. https://doi.org/10.1073/pnas.2120510119

[7] Jonathan Charteris-Black. 2011. Persuasion, Speech Making and Rhetoric. In *Politicians and Rhetoric: The Persuasive Power of Metaphor*, Jonathan Charteris-Black (Ed.). Palgrave Macmillan UK, London, 1–27. https://doi.org/10.1057/9780230319899_1

[8] Woojin Chung and Jeonghoon Kim. 2025. Exploiting Vocabulary Frequency Imbalance in Language Model Pre-training. https://doi.org/10.48550/arXiv.2508.15390 arXiv:2508.15390 [cs].

[9] Katherine Clayton, Nicholas T. Davis, Brendan Nyhan, Ethan Porter, Timothy J. Ryan, and Thomas J. Wood. 2021. Elite rhetoric can undermine democratic norms. *Proceedings of the National Academy of Sciences* 118, 23 (June 2021), e2024125118. https://doi.org/10.1073/pnas.2024125118

[10] Eurobarometer. 2023. *Media & News Survey 2023 - November 2023 - - Eurobarometer survey*. Technical Report. European Union. https://doi.org/10.2861/11595

[11] Bastian Herre. 2025. Democracy data: how sources differ and when to use which one. *Our World in Data* (June 2025). https://ourworldindata.org/democracies-measurement

[12] Bastian Herre. 2025. The 'Varieties of Democracy' data: how do researchers measure democracy? *Our World in Data* (April 2025). https://ourworldindata.org/vdem-electoral-democracy-data

[13] Günter Holtus and Michele Cortelazzo. 1985. Dal parlato al (tra)scritto: i resoconti stenografici dei discorsi parlamentari. In *Gesprochenes Italienisch in Geschichte und Gegenwart*. G. Narr, 86–118. https://www.uni-saarland.de/fileadmin/upload/fachrichtung/romanistik/W._Schweickard/Publikationen/Gesprochenes_Italienisch_in_Geschichte_und_Gegenwart.pdf Google-Books-ID: NfwlAAAAMAAJ.

[14] Zongxi Li, Xianming Li, Yuzhang Liu, Haoran Xie, Jing Li, Fu-lee Wang, Qing Li, and Xiaoqin Zhong. 2023. Label Supervised LLaMA Finetuning. https://doi.org/10.48550/arXiv.2310.01208 arXiv:2310.01208 [cs].

[15] Philipp Lorenz-Spreen, Lisa Oswald, Stephan Lewandowsky, and Ralph Hertwig. 2023. A systematic review of worldwide causal and correlational evidence on digital media and democracy. *Nature Human Behaviour* 7, 1 (Jan. 2023), 74–101. https://doi.org/10.1038/s41562-022-01460-1

[16] Aamer Madhani and Jill Colvin. 2021. A farewell to @realDonaldTrump, gone after 57,000 tweets. https://apnews.com/article/twitter-donald-trump-ban-cea450b1f12f4ceb8984972a120018d5

[17] Bron Maher. 2024. Twice as many Brits got 2024 election news from TV as from social media, Ofcom finds. https://pressgazette.co.uk/news/ofcom-2024-election-news-sources-television-social-media-working-class/

[18] Aurelia Mohrhoff. 1987. Dalla lingua del Parlamento alla lingua del parlamentare. In *Aurelia Mohrhoff, Dalla lingua del Parlamento alla lingua del parlamentare*. Roma: Camera dei Deputati, 145–156. https://bpr.camera.it/bpr/allegati/show/3815_828_t

[19] Benjamin I. Page and Robert Y. Shapiro. 1984. Presidents as Opinion Leaders: Some New Evidence. *Policy Studies Journal* 12, 4 (June 1984), 649–662. https://doi.org/10.1111/j.1541-0072.1984.tb00480.x

[20] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2022. Robust Speech Recognition via Large-Scale Weak Supervision. https://doi.org/10.48550/arXiv.2212.04356 arXiv:2212.04356 [eess].

[21] Paulo Rita, Nuno António, and Ana Patrícia Afonso. 2023. Social media discourse and voting decisions influence: sentiment analysis in tweets during an electoral period. *Social Network Analysis and Mining* 13, 1 (March 2023), 46. https://doi.org/10.1007/s13278-023-01048-1

[22] Jan Sawicki, Maria Ganzha, and Marcin Paprzycki. 2023. The State of the Art of Natural Language Processing—A Systematic Automated Review of NLP Literature Using NLP Techniques. *Data Intelligence* 5, 3 (Aug. 2023), 707–749. https://doi.org/10.1162/dint_a_00213

[23] Bruno Castanho Silva and Sven-Oliver Proksch. 2022. Politicians unleashed? Political communication on Twitter and in parliament in Western Europe. *Political Science Research and Methods* 10, 4 (Oct. 2022), 776–792. https://doi.org/10.1017/psrm.2021.36

[24] Sara Tonelli, Rachele Sprugnoli, and Giovanni Moretti. 2019. Prendo la Parola in Questo Consesso Mondiale:. (2019). https://ceur-ws.org/Vol-2481/paper71.pdf

[25] Alessandro Vinci. 2025. Meloni leader anche sui social, 3,3 milioni di nuovi follower nel 2025: staccato Salvini. Schlein prima per engagement. https://www.corriere.it/tecnologia/25_dicembre_17/meloni-leader-anche-sui-social-3-3-milioni-di-nuovi-follower-nel-2025-staccato-salvini-schlein-prima-per-engagement-431d8cda-a6e3-4d04-880b-922639075xlk.shtml Section: Meloni leader anche sui social, 3,3 milioni di nuovi follower nel 2025: staccato Salvini. Schlein prima per engagement.

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